

Champion statistic

When referring to units (such as champions) a **statistic** measures the power of an unmodified basic attribute or capability. For example: how durable they are; how fast they are able to move; how quickly they are able to perform basic attacks; etc. A **base** effect (e.g. how much damage an ability deals) that increases in strength by a statistic is said to "**scale with**" or "scale off of" that statistic. Scaling type can either directly or indirectly influence the champion's capabilities, and often heavily correlates to the champion's design. Statistics define a champion's **numerical** strength at various stages of the game or a fight, but not necessarily their overall strength — nevertheless, big differences in statistics also mean a substantial difference in the way different champions and champion classes interact with each other on the battlefield, as well as influence the game.

Statistics may be enhanced by:

[Runes](#)

[Items](#)

[Buffs](#)

Various bonuses from [abilities](#)

And **leveling up**:

All champions are able to gain  **experience points (XP)**. Experience is a unit that allows champions to obtain **levels** at certain thresholds, in order to naturally increase certain statistics. The amount of experience points required to level up increases with each level, and is the exact same for all champions. Some statistics are given to champions by the game rules, and are unique to every champion:

Base statistics are those statistics that have unmodified values at level 1.

Growth statistics are those increases to the base statistics that are gained explicitly by leveling up.

Increases that are not gained purely through levels (meaning they are not base statistics, and may come from any source other than growth), are called **bonus** statistics. *The exception is attack speed growth, which is gained per level, but the statistic counts toward "bonus attack speed".*

In the HUD, if there are any **bonuses** present in a statistic, the number will be highlighted in yellow; otherwise the value will remain grey.

Types

There are 23 types of champion statistics divided into 3 different categories: Defensive, Offensive, and Utility. In the list below, only the basic attribute of each statistic is presented. *In League of Legends, modified attacks, abilities, items, and runes, may freely scale off of **any** and/or **multiple** statistics (and sometimes other effects), whether in their damage, defense, cooldown, or any other attribute.*

Defensive

-  **Health [HP]**: A champion dies when their health is reduced to zero. Some abilities and effects may scale off of your own, on your ally's, or on a target enemy's: current, bonus, missing, or maximum health.
-  **Health regeneration [HP5]**: The amount of **health** a champion passively restores per 5 seconds.
-  **Heal and shield power**: Increases the strength of **healing** or **shielding** effects on yourself or allies. *See page for related attributes, mechanics, and exceptions.*
-  **Armor [AR]**: Reduces (mitigates) the amount of **physical damage** taken, thus increasing a champion's effective health against physical damage.
-  **Magic resistance [MR]**: Reduces (mitigates) the amount of **magic damage** taken, thus increasing a champion's effective health against magic damage.
-  **Tenacity**: Reduces the **duration** of most hostile **crowd control** effects/disables by a percentage. Stacking multiple sources of tenacity has diminishing returns. Disables cannot be reduced to under 0.5 seconds.
-  **Slow resist**: Reduces the **strength** of incoming hostile **slows** (movement speed reductions). The reduction is a percentage of the slow's strength.

Offensive

-  **Attack speed [AS]**: The number of basic attacks your champion is allowed to perform per second. *See page for other relevant attributes.*
-  **Attack damage [AD]**: One of the two main offensive statistics, along with Ability Power. **Unmodified** [basic attacks](#) deal exactly this amount of damage.
-  **Ability power [AP]**: One of the two main offensive statistics, along with Attack Damage.
-  **Critical strike chance [Crit%]**: Denotes the chance that a basic attack will critically strike. Certain abilities can also critically strike for modified damage.
-  **Critical strike damage**: Denotes the damage dealt when a basic attack critically strikes. Abilities that can critically strike usually also benefit from critical strike damage.
-  **Armor penetration**: When applying **physical damage** to an enemy, ignores a part of their **armor** in the damage calculations. Armor penetration can be either flat (*lethality*) or percentage-based. The target's armor is unchanged.
-  **Magic penetration**: When applying **magic damage** to an enemy, ignores a part of their **magic resistance** in the damage calculations. Magic penetration can be either flat or percentage-based. The target's magic resistance is unchanged.
-  **Life steal**: How much **health** a champion restores, as a percentage of the post-mitigation damage dealt by basic attacks and **on-hit** effects.
-  **Physical vamp**: How much **health** a champion restores, as a percentage of all **physical damage** dealt. Physical vamp is only 33% as effective when dealing **area damage** or pet damage.
-  **Omnivamp**: How much **health** a champion restores, as a percentage of all damage dealt. Omnivamp is only 20% as effective against non-champions.

Utility

All **haste**, but more commonly  **ability haste [AH]**: Reduces the [cooldown](#) of abilities by a percentage. Stacking ability haste itself has diminishing returns, but not in terms of cooldowns and casting.

Ability haste is just one of several forms of haste that a champion can obtain, but it is the only one actually displayed on a champion's statistics panel.

Resource refers to a mechanic meant to gate and be spent on ability casting. Each resource has a maximum pool or amount, and may or may not regenerate passively. Resource and resource regeneration are counted as two distinct and innate statistics for **all** champions, regardless of whether they use a certain resource type or not. Below are listed two prevalent resources that interact with resource-generic effects, unlike all others. *For all non-mana-users, see [here](#).*

 **Mana [MP]**: **Mana** is a longer-term resource that replenishes fairly slowly, but can be improved into great supply. Some effects may scale off a portion or all of your mana.

 **Mana regeneration [MP5]**: The amount of **mana** a champion passively restores per 5 seconds.

 **Energy**: **Energy** is a shorter-term resource that replenishes fairly quickly, but comes in limited supply compared to mana. All champions who use energy also have at least one means by which they are able to restore a flat amount of energy.

 **Energy regeneration**: The amount of **energy** a champion passively restores. Energy regeneration rate is a base of 50 energy per 5 seconds for all energy users, but it can be increased through a few means.

 **Attack range**: The maximum distance that a basic attack can be initiated from. Attack range is measured in *units* and refers to the distance from the source's edge to the target's edge.

When referring to **range** in general — abilities, item actives, and summoner spells' maximum distance and/or dimensions (where applicable) can also be included. Ability ranges have different targeting conditions.

 **Movement speed [MS]**: How quickly a champion moves across the map, measured in *units* per second.

*Note: "**Units**", arbitrarily named, are the basic units of game-distance, and they are used to reference all lengths and attack/ability speeds.*

 **Gold generation**, which is the rate at which a champion passively generates gold, measured per 10 seconds.

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Increasing Statistics

There are eight **base** statistics that may increase through **growth**: [attack damage](#), [attack speed](#), [armor](#), [magic resistance](#), [health](#), [health regeneration](#), [mana](#), and [mana regeneration](#).

The amount of those statistics gained by leveling up increases with each level (it is non-linear):

- When leveling up from level 1 to level 2, the champion will gain 72% of the statistic's growth.
- When leveling up from level 9 to level 10, the champion will gain 100% of the statistic's growth.
- When leveling up from level 17 to level 18, the champion will gain 128% of the statistic's growth.

To calculate a champion's statistics at a specific level, the following formula is used:

$$\text{Statistic} = \text{base} + \text{bonus} + g \times (n - 1) \times (0.7025 + 0.0175 \times (n - 1))$$

base = initial statistic value
bonus = bonus statistic from any source (runes, items, buffs, etc.)
g = growth statistic
n = current champion level
(n - 1) = total number of level ups

The **growth statistic** value is the rate at which each individual statistic increases. Each statistic has a separate growth statistic, and it cannot be changed (with the exception of when  Gnar and  Mega Gnar change forms).

EXAMPLE:

 Alistar starts at level 1 with **685 health** (*health base statistic*). His **health growth** statistic is **120**.
At level 2, applying the equation (base = 685, g = 120, n = 2) his **total health** is calculated as: **771.4**
This is displayed in game as a rounded value **771**.

Four statistics that deviate from the typical design are [gold generation](#), [critical strike damage](#), [range](#), and [movement speed](#).

- All champions on the same map start the game with the same [gold generation](#).
- All champions start with [175% base critical damage](#) (even  Ashe,  Jhin,  Yasuo, and others despite their innate passives).
- Every champion has unique [base range](#) and [movement speed](#) that never increases naturally. Natural increase means explicitly from leveling up itself—*innate passives such as*  Tristana's  Draw a Bead or  Cassiopeia's  Serpentine Grace, despite scaling per level, are not considered to be growth statistics.

All other statistics have a natural value of zero and must be increased through **bonus statistics**. The ways a champion may increase their statistics are varied and include [runes](#), [items](#), and [buffs](#) and [auras](#). Some champion abilities increase that champion's statistics, either passively ( Nasus'  Soul Eater passively increases his [life steal](#)), actively when cast ( Jax's  Grandmaster-At-Arms briefly increases his [armor](#) and [MR](#) after being cast), or conditionally if a condition is satisfied ( Nocturne's  Shroud of Darkness, for example, doubles the ability's passive bonus [attack speed](#), if he successfully blocks a spell with the spell shield).

Attack speed calculations

[Attack speed](#) uses different types of values, instead of simply a base and a growth amount:

Base attack speed
Attack speed ratio, which may be the same value as the base attack speed or a different one, depending on the champion
Bonus attack speed growth, as a percentage (%)
Bonus attack speed from other sources, as a percentage (%)

Because of the facts that:

- The base and ratio values are functionally unchanging, and the ratio is a multiplier for all attack speed gained
- Attack speed increases in percentages
- Attack speed gained through levels is considered different from attack speed gained through any other source, but they are both added together as "bonus attack speed"
The above statistic formula must be adjusted.

$$\text{Total Attack Speed} = \text{baseAs} + [\text{bonusAs} + g \times (n - 1) \times (0.7025 + 0.0175 \times (n - 1))] \times \text{AsRatio}$$

baseAs = base attack speed
AsRatio = attack speed ratio, if different from base attack speed; otherwise, replace with baseAs
g = percent bonus attack speed growth, gained explicitly from leveling up
bonusAs = sum of any percent bonus attack speed gained explicitly from any source [except leveling up](#)
n = current champion level
(n - 1) = total number of level ups

Example: Using  Volibear at level 3 with his passive fully stacked

Volibear's base attack speed is 0.625
Volibear's attack speed ratio is 0.7
Volibear's attack speed growth is 2%
Volibear's  passive grants him 5% bonus attack speed stacking up to 5 times, for a total of 25%

Ignoring all sources of bonus attack speed except his passive, Volibear's attack speed at level 3 and fully stacked The Relentless Storm will be:

$$\text{baseAs} + [\text{bonusAs} + g \times (n - 1) \times (0.7025 + 0.0175 \times (n - 1))] \times \text{AsRatio}$$

$$0.625 + [0.25 + 0.02 \times (3 - 1) \times (0.7025 + 0.0175 \times (3 - 1))] \times 0.7$$

$$0.625 + [0.25 + 0.0295] \times 0.7$$

$$0.625 + [0.2795] \times 0.7$$

$$0.625 + 0.19565$$

$$\text{Total Attack Speed} = 0.82065$$

Growth statistic calculations

To calculate the amount of statistic gained per level, use the following formula, derived from the growth term of the statistics formula:

$$\text{Stat Increase} = g \times (0.65 + 0.035 \times \text{New Level})$$

(Where g = growth statistic)

The number in the bracket determines the percentage of increase that happens at each level up.

At level 2, that is 0.65 + 0.035× 2 = 0.72 or 72%.

At level 10, that is 0.65 + 0.035× 10 = 1.00 or 100%.

At level 18, that is 0.65 + 0.035× 18 = 1.28 or 128%.

By level 18, a champion will have gained 1700%, or 17 times, the growth statistic. Their total statistic would be BaseStat + 17 × GrowthStat

Notes

Statistics having a "base" value was established in patch V4.20. These are the values **at level 1**.^[1] These values can also be verified by entering a game with no runes and viewing the champion statistics panel.

The same patch changed stat growth from linear to quadratic (growth increases with increasing level).

In the champion stats panels:

Attack damage, ability power, armor, magic resistance, attack speed, movement speed, ability haste and critical strike chance can be found in the basic (or default) stats menu (the player can show or hide the panel by clicking on the panel's button).

Health regeneration, mana regeneration, heal and shield power, armor penetration, magic penetration, life steal, physical vamp, omnivamp, attack range, and tenacity can be found in the advanced player stats menu (default C to toggle and Ctrl + C to cycle).

For each statistic, if its value is written in white then only the base amount is present, and if its value is is written in bold yellow then it has been increased from the base amount through bonuses. By hovering the cursor over almost any statistic, a tooltip can be accessed that includes extra information about the statistic and/or more precise values. Each stat can be hovered to show a tooltip that includes a brief description of a statistic, and total, base, and bonus values, in the format total (base + bonus). The base value is colored blue, and the bonus amount is colored green. If there is no bonus amount then the total value is colored blue, and if there are any bonuses then the total value is colored green. This coloring scheme applies to all visible stats, with the exceptions of armor penetration and magic penetration which are always written in blue (an inconsistency, but purely cosmetic nonetheless).

Armor and magic resistance also include the resulting damage reduction in their respective tooltips.

Health regeneration and resource regeneration are presented together as "Regen" and in the format Health regen. | Resource regen..

Armor penetration and magic penetration are presented in the format Flat | Percentage.

Total critical damage is not shown to the player. Within the critical strike chance tooltip, the value of 75% base critical damage is always listed regardless of any bonuses. The presented information also does not consider any other forms or values for critical strikes.

Basic stats, which include health, resource, attack damage, ability power, armor, magic resistance, attack speed, movement speed, ability haste and critical strike chance (along with buffs and debuffs) are displayed on the target frame, when a unit is selected. Hovering the cursor over the health line or the resource line in the HUD additionally displays health regeneration and resource regeneration in amount per second, as opposed to the regular amount per 5 seconds. The player can also hover their cursor over their own health and resource bar to view these values per second.

Infinite Stat Stacking

The following 15 champions have an ability that infinitely stacks an effect: Aurelion Sol, Bard, Bel'Veth, Cho'Gath, Draven, Kindred, Nasus, Senna, Shyvana, Sion, Smolder, Swain, Sylas, Thresh, and Veigar.

Aurelion Sol can infinitely augment the potency of his abilities by collecting Stardust stacks.

Bel'Veth can infinitely stack her attack speed with Lavender stacks.

Cho'Gath can infinitely stack its health with Feast stacks.

Draven can infinitely gain Adoration stacks, assuming he does not die.

Kindred can infinitely stack the bonus attack speed gained from Dance of Arrows which scales based on how many Marks they have.

Nasus can infinitely stack the bonus damage of his Siphoning Strike by using it to kill enemies.

Senna can infinitely stack her bonus attack damage, bonus range, and life steal by converting critical strike chance beyond the cap from Mist stacks.

Shyvana can infinitely stack her resistances from Fury of the Dragonborn.

Sion can infinitely stack his health with Soul Furnace's passive stacks.

Smolder can infinitely stack the bonus magic damage on his basic abilities, aswell as the amount of bolts and true damage from Super Scorchers Breath by collecting Dragon Practice stacks.

Swain can infinitely stack his health with Soul Fragment stacks.

Sylas can infinitely stack his health from Hijacking an enemy Cho'Gath's Feast and gaining stacks of the ability as they are permanent.

Thresh can infinitely stack his ability power and armor from Soul stacks.

Veigar can infinitely stack his ability power from Phenomenal Evil Power stacks.

While not a scaling, Yuumi can also infinitely stack friendship on ally champions with Feline Friendship.

Trivia

Passive gold generation, slow resist, and total critical strike damage are the only statistics that have to be calculated by the player during the game, and are not explicitly displayed anywhere in the game's menus.

Heal and shield power was added as a functional statistic as a passive item effect in V6.13, standardized in V11.8 as a proper statistic, but only added to the stats panel in V14.10, almost eight years after its release.

Lethality was implemented as a standard statistic in V14.9. Before this, it would be handled through scripts.

Some statistics' effectiveness may also be relevant when combined with other statistics that both interact with a certain mechanic. For example, attack damage directly benefits the raw amount of health you gain back from life steal; both interact with the mechanic of basic attacks; and there is always an equilibrium of effectiveness among such statistics, depending on their availability at the time.

See also

Base champion statistics

Champion statistics at level 18

References

- Patch 4.20 Notes (https://web.archive.org/web/0/http://oce.leagueoflegends.com/en/news/game-updates/patch/patch-420-notes#patch-stats-gained-per-level)

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